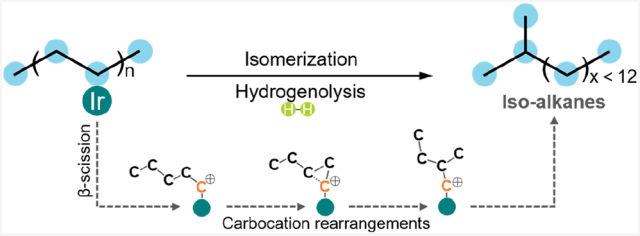
JACS
JOURNAL OF THE AMERICAN CHEMICAL SOCIETY
pubs.acs.org/JACS
Article

Confined Isomerization Enables Solar-Thermal Upcycling of Polyolefins into High-Octane Gasoline

Chengyang Feng, Miao Hu, Shouwei Zuo, Bo Li, Yuanfu Ren, Zhi-Peng Wu, Jumanah Alharbi, Jun Luo, Wan-Lu Li, Nikos Hadjichristidis, Magnus Rueping, and Huabin Zhang*

Cite This: J. Am. Chem. Soc. 2026, 148, 33433−33442
Read Online
ACCESS | Metrics & More | Article Recommendations | Supporting Information

ABSTRACT: Upcycling polyolefin waste into liquid fuels through hydrogenolysis holds great promise for advancing a circular carbon economy. However, extensive skeletal isomerization crucial for high-octane gasoline production remains a fundamental and unresolved limitation. Herein, we introduce a self-confined adsorption-configuration strategy that enables isomerization during polyolefin hydrogenolysis. By enforcing single-point adsorption on isolated Ir sites, β-scission intermediates gain the geometric freedom necessary for carbocation rearrangements, ultimately enabling the direct production of high-octane gasoline under solar-thermal conditions. In addition to promoting internal C−C bond rearrangement, the confined adsorption mode suppresses terminal cracking pathways. The Ir SA/WO_3 catalyst achieves 92% conversion of polyolefin waste to liquid hydrocarbons while nearly eliminating methane production. Overall, this work advances a practical route for upcycling waste plastics into valuable energy products, contributing to a more closed-loop and carbon-efficient anthropogenic carbon cycle.

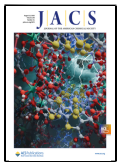


1. INTRODUCTION

By 2050, the cumulative amount of plastic waste is projected to exceed 25 billion tons, with polyolefins, primarily polyethylene (PE) and polypropylene (PP), accounting for nearly 70% of the total and representing the dominant component of municipal solid plastic waste.[1−4] Current mechanical recycling strategies impose stringent purity requirements on waste plastics, and the recycled materials usually suffer from severe property degradation, rendering them unsuitable for multiple recycling cycles.[5−9] Consequently, the vast majority of discarded plastics ultimately end up in landfills, incinerators, or natural ecosystems, leading to an ever-worsening global environmental crisis and massive resource losses (Figure S1).[10−12] Transforming polyolefin waste into valuable chemicals or renewable liquid fuels has therefore emerged as a promising pathway toward a circular carbon economy.[13,14] This approach, termed plastic upcycling, envisions repurposing waste polymers as an alternative carbon source to crude oil, effectively converting plastic refuse into an urban oilfield and unlocking a sustainable route for both waste mitigation and resource recovery.[15−24]

Common strategies for plastic upcycling include thermal pyrolysis and catalytic cracking, although these approaches typically suffer from high energy consumption, broad product distributions, and severe catalyst coking.[25,26] In contrast, catalytic hydrogenolysis can convert polyolefin waste into liquid hydrocarbon fuels under comparatively mild conditions, usually below 400 °C, while significantly reducing coke formation.[13] This makes hydrogenolysis an appealing route for sustainable plastic valorization, particularly in view of the rapid expansion of green hydrogen production. Despite these advantages, several critical challenges remain. One major limitation is that conventional hydrogenolysis catalysts inherently favor the formation of highly saturated linear *n*-alkanes, which increases the difficulty and cost of downstream refining. Although acidic zeolites and metal-acid bifunctional catalysts can promote skeletal isomerization, they often suffer from overcracking, coke formation, diffusion limitations for bulky polyolefin chains, and sensitivity to impurities. Therefore, achieving extensive skeletal isomerization over nonacidic or weakly acidic oxide catalysts remains a key challenge.[27−30] Another challenge arises from the behavior of conventional metal and metal oxide catalysts. These catalysts tend to promote terminal C−C bond scission along the polyolefin backbone, generating large amounts of low-value methane.[31−34] Both the insufficient isomerization and the high methane selectivity can be traced to the same mechanistic origin, which is the multipoint adsorption configuration of long-chain polyolefins on traditional catalytic surfaces (Figure

Received: April 21, 2026
Revised: July 20, 2026
Accepted: July 23, 2026
Published: July 29, 2026



ACS Publications
© 2026 American Chemical Society

https://doi.org/10.1021/jacs.6c07894
J. Am. Chem. Soc. 2026, 148, 33433−33442

Downloaded from pubs.acs.org/jacsat/article-pdf/148/31/33433/6330499/jacs.6c07894.pdf by INDIAN INST OF TECH DELHI- IIT user on 30 September 2026

1a). When the polymer chain is immobilized through multiple surface interactions, the carbocation rearrangements that

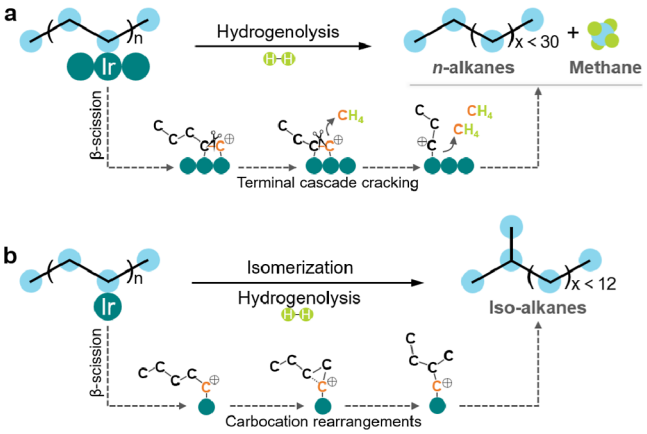


Figure 1. Overview of solar thermal hydrogenolysis of polyolefin waste plastics. (a) Hydrogenolysis induced by multisite adsorption produces *n*-alkanes and methane. (b) Hydrogenolysis induced by single-site adsorption, self-confinement promotes internal cracking and isomerization.

enable skeletal isomerization are kinetically hindered. The transition states required for rearrangement are rapidly terminated by hydrogenation before they can develop, causing the chain to crack primarily into linear light alkanes. On the other hand, the multipoint adsorption configuration is prone to triggering terminal cascade reactions, leading to continuous terminal hydrogenation to produce methane.[28,31,35,36] These mechanistic insights indicate that tuning the nature of the catalytic sites in order to modify the adsorption configuration of polyolefin chains provides a promising route to simultaneously suppress terminal cracking and enhance internal C−C scission as well as skeletal isomerization.

Herein, we present a self-confined strategy for the hydrogenolysis of polyolefin waste using a WO_3 catalyst anchored with isolated Ir single atoms (Ir SA/WO_3), which enables the direct conversion of polyolefin plastics into high-octane gasoline under mild reaction temperatures of 200 to 350 °C while simultaneously suppressing methane formation. The isolated Ir sites promote a single-point adsorption configuration on the interior carbons of the polymer chain. This adsorption mode disrupts the conventional pathway of terminal C−C bond scission and simultaneously endows the β-scission intermediates with sufficient geometric flexibility to undergo carbocation rearrangements (Figure 1b). As a result, the reaction pathway shifts toward internal C−C cleavage and extensive skeletal isomerization, enabling the direct formation of highly branched gasoline-range hydrocarbons. Furthermore, by integrating the Ir SA/WO_3 catalyst with a custom-designed solar-thermal reactor, we establish a solar-driven system for the upcycling of plastic waste. Under natural sunlight, this system continuously produces commercial-grade recycled gasoline with an experimental octane value of approximately 90, demonstrating its strong potential for sustainable plastic valorization.

2. RESULTS AND DISCUSSION

2.1. Synthesis and Characterization of Catalysts

The Ir/WO_3 catalysts were synthesized via a facile impregnation-calcination method, in which the Ir loading amount was adjusted to control the dispersion state of Ir species, resulting in the formation of single-atom Ir (Ir SA/WO_3) and nanoparticle Ir (Ir NP/WO_3) catalysts (Table S1). The prepared Ir/WO_3 catalysts (including Ir SA/WO_3 and Ir NP/WO_3) exhibited the orthorhombic phase of WO_3. No diffraction peaks corresponding to Ir species were observed in the X-ray diffraction (XRD) patterns (Figure S2). Electron microscopy was conducted to analyze the morphology and dispersion state of Ir species in the Ir/WO_3 samples. High-resolution TEM images of Ir NP/WO_3 revealed that Ir species mainly exist in the form of nanoparticles (Figure S3), with sizes below 5 nm (Figure S4). In contrast, no Ir nanoparticles were observed on the surface of Ir SA/WO_3 (Figure S5). Combined with the high-angle annular dark-field scanning transmission electron microscopy (HAADF-STEM) images, randomly dispersed atomic-scale bright spots with lower contrast were identified on the surface of Ir SA/WO_3 (Figures 2a and S6), which can be attributed to isolated Ir atoms anchored on the WO_3 surface. Energy-dispersive X-ray spectroscopy (EDS) elemental mapping further confirmed that these isolated Ir species are uniformly distributed (Figure 2b).

Further, the Fourier transform extended X-ray absorption fine structure (EXAFS) spectra revealed that Ir NP/WO_3 exhibited a predominant peak at 2.11 Å in the first coordination shell, which is very close to the Ir−Ir scattering path observed in Ir foil. In contrast, Ir SA/WO_3 showed a distinct feature at 1.74 Å, corresponding to the Ir−O coordination similar to that in IrO_2, without any detectable Ir−Ir or Ir−O−Ir contributions, thereby excluding the presence of Ir-related nanoparticles or clusters (Figure 2c). Quantitative EXAFS fitting further resolved the first coordination shell of Ir SA/WO_3 as an Ir−O_5 configuration, consisting of four basal Ir−O bonds and one slightly elongated apical Ir−O bond, providing a structural basis for the DFT model (Figure S7 and Table S2). Moreover, the wavelet transform (WT) contour plots further demonstrated that the coordination environments of Ir NP/WO_3 and Ir SA/WO_3 closely resemble those of Ir foil and IrO_2, respectively (Figure S8), confirming the successful synthesis of both nanoparticle and single-atom Ir catalysts. The X-ray absorption near-edge structure (XANES) and X-ray photoelectron spectroscopy (XPS) results (Figures 2d and S9) further support this conclusion, indicating that the Ir species in Ir SA/WO_3 exist predominantly in an oxidized state, whereas those in Ir NP/WO_3 exhibit a more metallic character.

2.2. Performance Evaluation for Polyolefin Solar Thermal Upcycling

Solar-thermal hydrogenolysis of polyolefin plastics was conducted in a custom-designed reactor system consisting of a parabolic concentrator and a high-pressure autoclave coated with an α-Fe_3O_4 photothermal layer, enabling rapid photothermal response and efficient solar-to-thermal energy conversion (Figure S10). For a typical reaction, low-density polyethylene (LDPE) and the catalyst were mixed at a mass ratio of 10:1, and the mixture was loaded into the autoclave. The reactor was purged with hydrogen and then pressurized to 10 bar. Outdoor experiments were typically carried out for 6 h (10:00−16:00, Figure S11), during which the reactor temper-

Downloaded from pubs.acs.org/jacsat/article-pdf/148/31/33433/6330499/jacs.6c07894.pdf by INDIAN INST OF TECH DELHI- IIT user on 30 September 2026

https://doi.org/10.1021/jacs.6c07894
J. Am. Chem. Soc. 2026, 148, 33433−33442

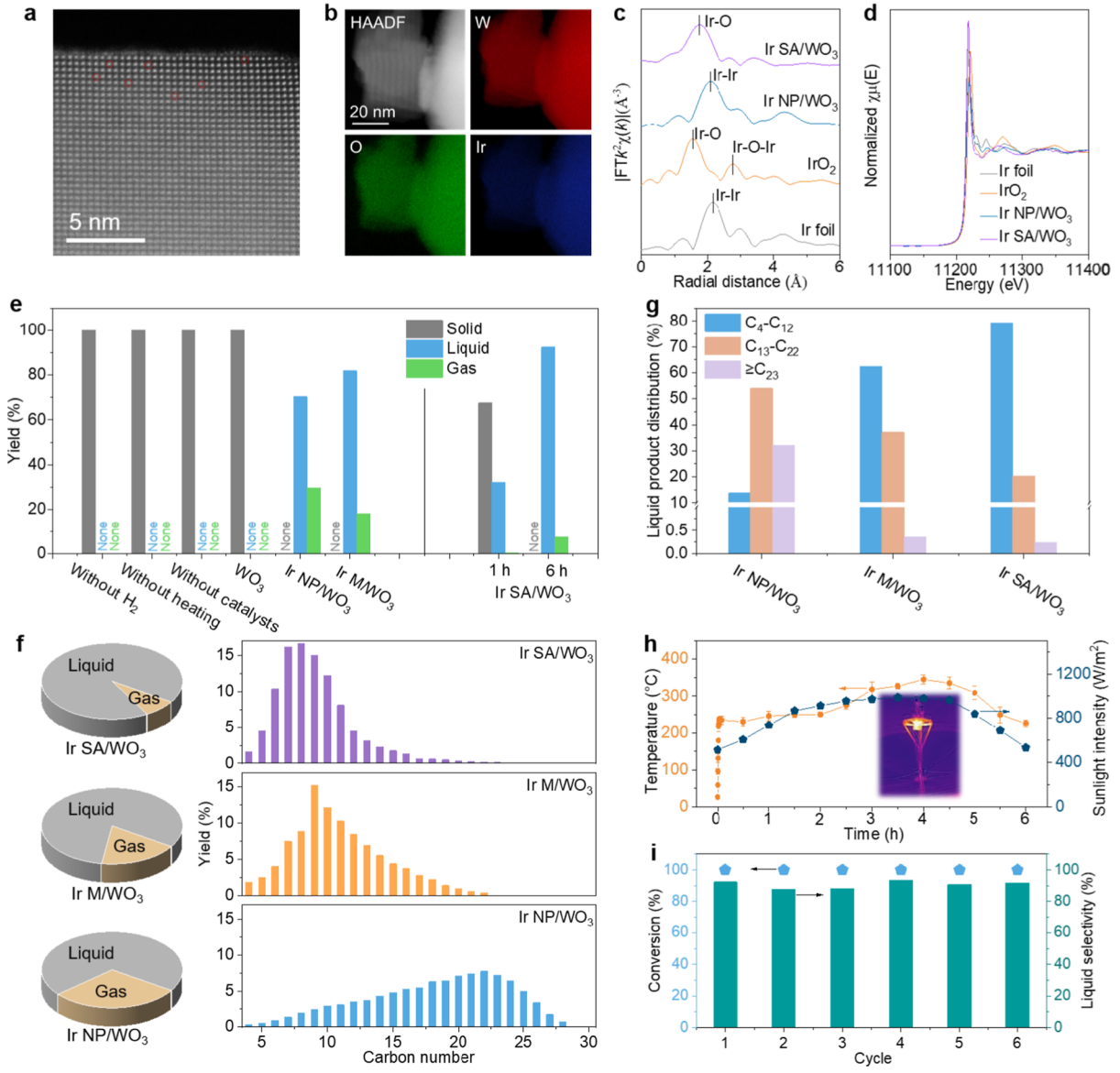


Figure 2. (a) High-magnification high-angle annular dark-field scanning transmission electron microscopy (HAADF-STEM) images of Ir SA/WO_3. (b) Energy dispersive spectroscopy elemental mapping of Ir SA/WO_3. (c) Ir L_3-edge Fourier-transformed extended X-ray absorption fine structure (EXAFS) spectra. (d) Ir L_3-edge X-ray absorption near edge structure (XANES) spectra. (e) Comparison of LDPE hydrogenolysis performance of prepared catalysts (solar thermal, 6 h). (f) Product composition and carbon number distribution of liquid products over Ir SA/WO_3, Ir M/WO_3, and Ir NP/WO_3. (g) Statistics on the proportion of gasoline in liquid products. (h) Recording of sunlight intensity and reactor temperature, the inset is a thermal imaging photo of the reactor during the reaction process. (i) Cycling stability of the Ir SA/WO_3 catalysts.

ature was maintained between 250 and 350 °C depending on solar intensity (Figure 2h). After the reaction, the system was cooled to room temperature to terminate the process, and the products were collected for analysis. Under all control conditions—namely, in the absence of hydrogen, at room temperature, or without a catalyst—no reaction occurred, confirming that the upcycling process is a solar-thermal-driven catalytic hydrogenolysis reaction (Figure 2e). Moreover, pristine WO_3 exhibited no activity toward LDPE cleavage, indicating that the Ir species serve as the active sites responsible for the catalytic hydrogenolysis of polyethylene.

To clarify whether the observed skeletal isomerization originates from conventional metal−acid bifunctional catalysis, we further evaluated the acidity of WO_3-based catalysts and oxide-supported Ir single-atom catalysts (Figures S12−S19). NH_3-TPD and pyridine-IR show that Ir SA/WO_3 possesses only weak Lewis acidity and no discernible Brønsted acid sites, unlike classical zeolite-based hydrocracking catalysts. Although TiO_2, ZrO_2, and Al_2O_3 exhibit stronger Lewis acidity than WO_3, their Ir single-atom catalysts do not surpass Ir SA/WO_3 in branching regulation. Moreover, Ru SA/WO_3 shows much stronger Lewis acidity and weak Brønsted acidity, yet affords a lower branching fraction than Ir SA/WO_3. These results indicate that the high isomerization selectivity cannot be simply attributed to conventional acid sites. Instead, isolated Ir sites play the key role in regulating C−C cleavage, skeletal rearrangement, and hydrogenation.

For the Ir NP/WO_3 catalyst, gaseous products accounted for up to 30% of the total yield, predominantly consisting of low-value CH_4 (Figure S20a). Notably, as the Ir species evolved from nanoparticles to isolated single atoms, the gas fraction decreased to below 8%, and CH_4 formation was almost

https://doi.org/10.1021/jacs.6c07894
J. Am. Chem. Soc. 2026, 148, 33433−33442

Downloaded from pubs.acs.org/jacsat/article-pdf/148/31/33433/6330499/jacs.6c07894.pdf by INDIAN INST OF TECH DELHI- IIT user on 30 September 2026

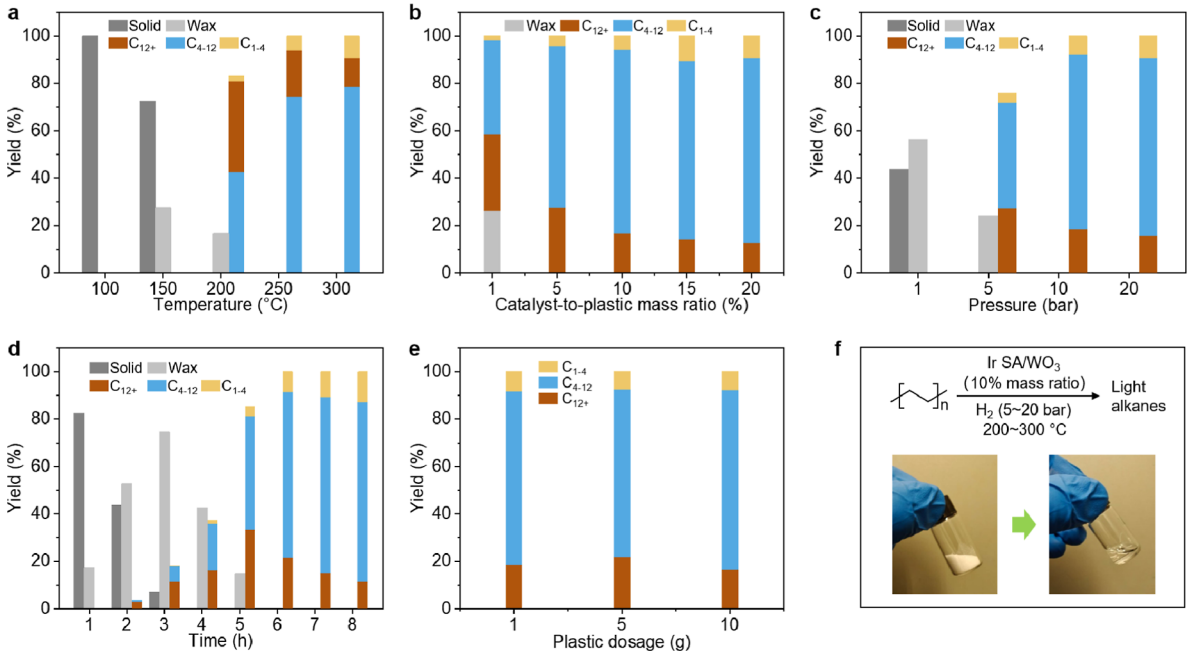


Figure 3. (a) Influence of temperature (1 g of LDPE, 100 mg of catalyst, electric heating, 6 h, 10 bar H_2). (b) Influence of catalyst loading (1 g of LDPE, 250 °C, electric heating, 6 h, 10 bar H_2). (c) Influence of H_2 pressure (1 g of LDPE, 100 mg of catalyst, 250 °C, electric heating, 6 h). (d) Product distribution changes with reaction time (1 g of LDPE, 100 mg of catalyst, 250 °C, electric heating, 10 bar H_2). (e) Influence of initial polyolefin dosage (10 wt % of catalyst, solar thermal, 6 h, 10 bar H_2). (f) Photographs of commercial LDPE plastic powder and its upgraded liquid-phase products.

completely suppressed (Figures S21a and 22a). Meanwhile, the composition of the liquid-phase products underwent a significant transformation. With the conversion of active sites from Ir nanoparticles to isolated Ir atoms, the carbon number distribution of the liquid hydrocarbons became narrower, and the average carbon chain length decreased (Figure 2f). The Ir SA/WO_3 catalyst exhibited a high selectivity of up to 80% toward C_{4-12} hydrocarbons (gasoline-range fraction, Figures 2g and S22b), whereas Ir NP/WO_3 mainly produced heavier diesel-range hydrocarbons (C_{12+}, Figures S20a and 21b). Based on the product distribution, the cleavage pattern of the LDPE backbone varies significantly depending on the geometric constraints of the Ir active sites (Figure S23). On Ir NP/WO_3, the Ir nanoparticles create compact adsorption domains that promote multipoint anchoring of polyolefin chains. This configuration facilitates sequential terminal C−C bond scission, leading directly to elevated methane formation. In contrast, the atomically dispersed Ir sites on Ir SA/WO_3 impose a strong self-confinement effect that restricts the hydrogenolysis polymer fragments to single-point adsorption. This confinement suppresses terminal chain-end cracking and instead channels the reaction toward stepwise internal C−C cleavage. The resulting geometric constraint markedly reduces methane production and favors the formation of short-chain hydrocarbons in the gasoline-range fraction.

The used catalysts were collected for recycling tests. After multiple consecutive runs, the catalytic activity remained stable, with negligible loss in both conversion and liquid-phase selectivity, demonstrating the excellent stability and durability of the catalyst (Figure 2i). Following the success of the model LDPE study, we further extended the solar-thermal hydrogenolysis process to other representative polyolefins. The Ir SA/WO_3 catalyst exhibited similarly high conversion and gasoline-range selectivity for LLDPE, HDPE, and PP (Figures S24−S26). These polymers collectively account for nearly 70% of real-world plastic waste, confirming the broad applicability and practical potential of the developed strategy for solar-driven catalytic upcycling of polyolefin plastics.

We further evaluated the effects of reaction temperature, catalyst loading, hydrogen pressure, and reaction time on the catalytic performance of Ir SA/WO_3. Temperature-controlled experiments revealed that carbon−carbon bond cleavage could be initiated at temperatures as low as 150 °C (Figure S29a); however, both the conversion and cracking degree were limited under these conditions, yielding only about 28% of wax-like products, with the remaining fraction largely retained as solid residues (Figures 3a and S27). When the temperature was raised to 200 °C, LDPE was completely converted, although considerable wax and C_{12+} liquid products were still observed. At temperatures above 250 °C, the residual wax fraction was fully cracked, giving a dominant liquid C_{4-12} hydrocarbon fraction with a narrowed liquid-phase carbon-number distribution. Further increasing the temperature to 300 °C slightly increased the liquid C_{4-12} yield, but also promoted the formation of C_{1-4} gaseous products, indicating the onset of overcracking at higher reaction severity. In addition, the fraction of branched hydrocarbons in the liquid products increased continuously from 200 to 300 °C, demonstrating that elevated temperature facilitates not only C−C bond cleavage but also skeletal isomerization over Ir SA/WO_3 (Figure S28). This optimal temperature window aligns well with the fluctuation range of solar-thermal input, highlighting the compatibility of the catalytic system with natural solar irradiation. The reaction rate exhibited a positive correlation with catalyst loading (Figure S29b). When the Ir SA/WO_3 dosage exceeded 5 wt %, complete conversion of the wax fraction was achieved. However, further increasing the loading beyond 15 wt % caused overcracking, resulting in a higher

https://doi.org/10.1021/jacs.6c07894
J. Am. Chem. Soc. 2026, 148, 33433−33442

Downloaded from pubs.acs.org/jacsat/article-pdf/148/31/33433/6330499/jacs.6c07894.pdf by INDIAN INST OF TECH DELHI- IIT user on 30 September 2026

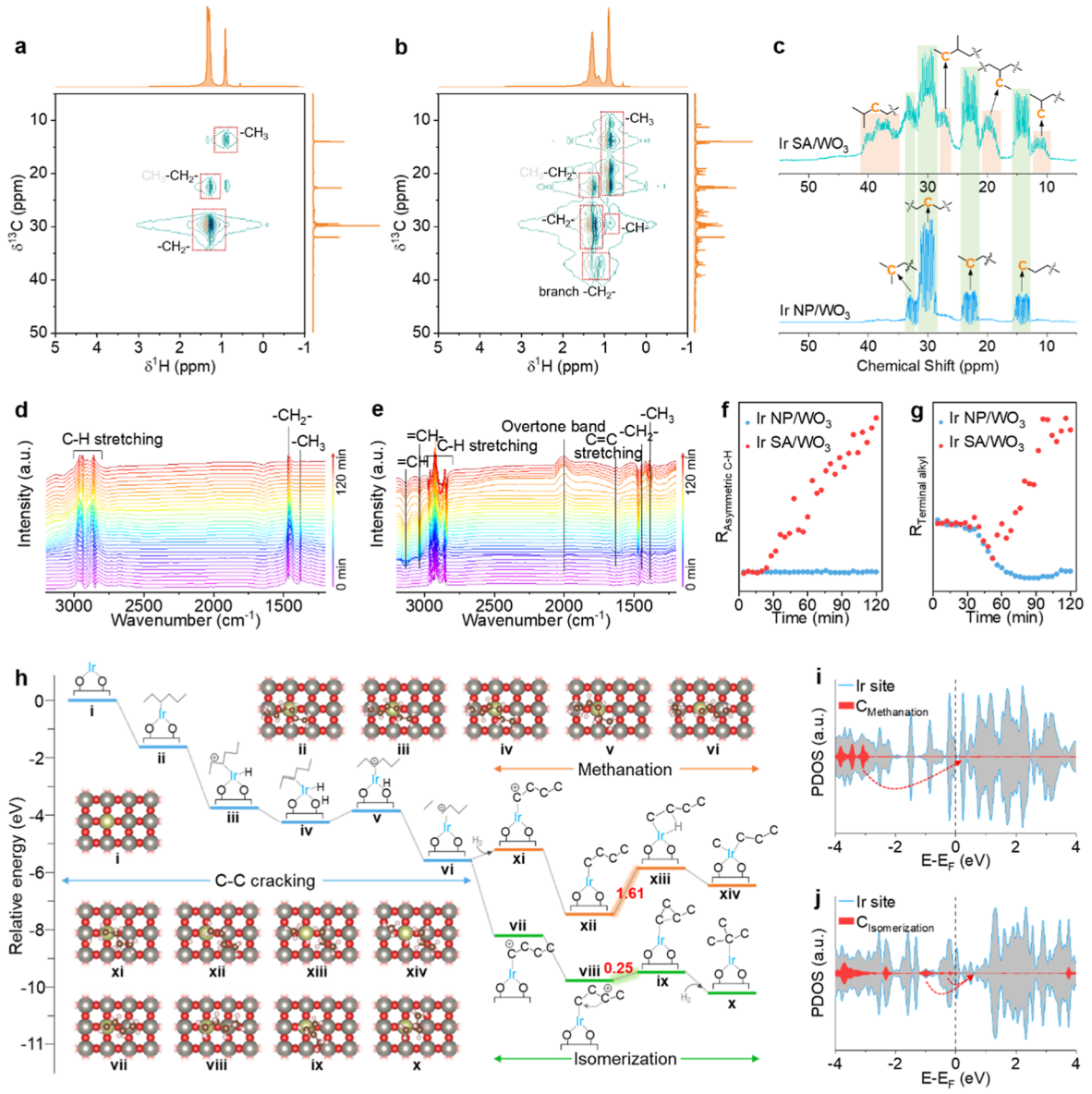


Figure 4. (a) Two-dimensional $^{1}H-^{13}C$ HSQC NMR spectra of liquid products formed from LDPE over Ir NP/WO_3. (b) Two-dimensional $^{1}H-^{13}C$ HSQC NMR spectra of liquid products formed from LDPE over Ir SA/WO_3. (c) ^{13}C NMR spectra of liquid products formed over Ir SA/WO_3 and Ir NP/WO_3. (d) In situ DRIFTS spectroscopies of LDPE hydrogenolysis over Ir NP/WO_3. (e) In situ DRIFTS spectroscopies of LDPE hydrogenolysis over Ir SA/WO_3. (f) Changes in the infrared intensity ratio associated with terminal alkyl groups, $R_{\text{Terminal alkyl}} = I_{(1378)}/I_{(1465)}$. (g) Changes in the infrared intensity ratio associated with asymmetric C−H stretching signals, $R_{\text{Asymmetric C-H}} = I_{(2923)}/I_{(2853)}$. (h) Gibbs free-energy diagrams for the reaction pathway over Ir SA/WO_3 catalyst. (i) PDOS of alkane-Ir site interaction in the rate-determining step of the methanation pathway. (j) PDOS of alkane-Ir site interaction in the rate-determining step of the isomerization pathway.

proportion of gaseous products (Figure 3b). Therefore, the catalyst loading was optimized at 10 wt %. Hydrogen pressure also appeared to significantly influence the hydrogenolysis of polyolefins (Figure 3c). However, kinetic analysis of the *n*-hexadecane hydrogenolysis revealed that, under sufficient hydrogen supply, the reaction rate was only weakly dependent on hydrogen pressure (Figure S29c). The primary effect of higher hydrogen pressure lies in maintaining an adequate hydrogen reservoir for the overall reaction rather than accelerating the intrinsic kinetics. The influence of reaction time on reaction efficiency and product distribution was more intuitive (Figure S29d). As the reaction proceeded, LDPE was first cracked into wax-like intermediates, followed by conversion into diesel-range hydrocarbons (C_{12+}), and subsequently into gasoline-range (C_{4-12}) and gaseous (C_{1-4}) products (Figure 3d). This sequential evolution of products is consistent with the proposed stepwise cracking mechanism, where polyethylene undergoes progressive chain scission toward shorter hydrocarbons over time. In addition, the scalability of the reaction system was preliminarily validated. By maintaining a constant catalyst-to-substrate ratio, the initial LDPE feed was gradually increased up to 10-fold. Under outdoor solar-thermal catalytic conditions, a comparable product distribution and gasoline-range selectivity were obtained, indicating that the catalytic performance can be effectively preserved upon scale-up (Figure 3e). This result highlights the robustness of the solar-thermal hydrogenolysis system and its potential for practical application in large-scale

https://doi.org/10.1021/jacs.6c07894
J. Am. Chem. Soc. 2026, 148, 33433−33442

Downloaded from pubs.acs.org/jacsat/article-pdf/148/31/33433/6330499/jacs.6c07894.pdf by INDIAN INST OF TECH DELHI- IIT user on 30 September 2026

plastic upcycling (Figure 3f). Furthermore, compared with recently reported polyolefin hydrocracking catalysts, Ir SA/WO_3 offers a balanced combination of high reactivity, methane generation inhibition, and enhanced skeletal isomerization (Figure S30), underscoring its potential for high-octane gasoline-range hydrocarbon production from polyolefin waste.

2.3. Geometrically Constrained Polyolefin Cleavage and Isomerization

Furthermore, the structure of the liquid-phase products was analyzed using nuclear magnetic resonance (NMR), in situ diffuse reflectance infrared Fourier transform spectroscopy (in situ DRIFTS), and gas chromatography−mass spectrometry (GC−MS). As shown in Figure 4a, the $^1H-^{13}C$ heteronuclear single-quantum coherence (HSQC) spectrum of the liquid-phase product obtained from polyethylene hydrogenolysis over Ir NP/WO_3 exhibits dominant correlations corresponding to linear alkane fragments. The cross-peaks at $\delta^{13}C \approx 30$ ppm/$\delta^1H \approx 1.3$ ppm and $\delta^{13}C \approx 14$ ppm/$\delta^1H \approx 0.9$ ppm are assigned to methylene ($-CH_2-$) and terminal methyl ($-CH_3$) groups, respectively, indicating minimal isomerization during chain scission. These results suggest that the Ir nanoparticles favor a classical β-scission pathway yielding linear alkanes. In contrast, the HSQC spectrum in Figure 4b corresponding to Ir SA/WO_3-catalyzed hydrogenolysis reveals not only the linear alkyl signals but also abundant cross-peaks associated with branched alkyl structures. The enhanced signals at $\delta^{13}C \approx 20-22$ ppm/$\delta^1H \approx 0.9-1.0$ ppm and $\delta^{13}C \approx 30-35$ ppm/$\delta^1H \approx 1.3-1.7$ ppm are attributed to methyl and methylene groups in branched alkanes, respectively. These observations indicate that the single-atom Ir sites facilitate isomerization during chain scission, promoting the formation of highly branched isomers. Notably, unlike previously reported results, the liquid-phase products obtained in this work contain no aromatic hydrocarbons (Figure S31), which renders the produced gasoline more environmentally benign and combustion-efficient. The absence of aromatics not only eliminates potential toxicity and carcinogenicity associated with aromatic compounds but also enhances the fuel's combustion stability and reduces soot formation during burning, thereby improving its overall quality and sustainability. The ^{13}C NMR spectra (Figure 4c) also display signals solely within the aliphatic region (10−50 ppm), with no evidence of olefinic or aromatic carbons, suggesting that the hydrogenolysis fully saturates the polymer chains. The product derived from Ir NP/WO_3 shows strong $-CH_2-$ signals and limited $-CH_3$ and $-CH-$, indicating predominant linear alkanes. In contrast, the Ir SA/WO_3-catalyzed products present significantly stronger $-CH-$ and $-CH_3$ signals, consistent with increased chain branching.

In situ diffuse reflectance infrared Fourier transform spectroscopy (DRIFTS) was employed to monitor the hydrogenolysis of polyethylene. On Ir NP/WO_3, the DRIFTS spectra show strong C−H stretching bands in the range of 2950−2850 cm^{-1} and deformation vibrations near 1460 cm^{-1} ($-CH_2-$) and 1375 cm^{-1} ($-CH_3$), indicating the formation of saturated linear alkanes (Figure 4d). These features remain largely unchanged over 120 min, suggesting minimal skeletal rearrangement during hydrogenolysis. In contrast, Ir SA/WO_3 shows not only similar alkyl vibrations but also additional bands at ∼3080 cm^{-1} (olefinic C−H stretching), ∼1950 cm^{-1} (overtone band), and ∼1600 cm^{-1} (C═C stretching) (Figure 4e). The emergence and intensification of these peaks over time point to the presence of unsaturated intermediates and branched products. This implies that single-atom Ir sites enable more extensive β-scission and skeletal rearrangements than Ir nanoparticles, consistent with the higher yield of branched alkanes and isomerized fragments observed in the corresponding NMR spectra. The time-dependent infrared intensity ratios reveal distinct structural evolution during polyethylene hydrogenolysis. For Ir SA/WO_3, $R_{\text{Terminal alkyl}}$ first decreases and then increases (Figure 4f), suggesting an initial consumption of terminal $-CH_3$ groups as the polymer chains adsorb and activate on the catalyst surface, followed by the generation of new terminal ends through successive β-scission events. In parallel, $R_{\text{Asymmetric C-H}}$ continuously rises, indicating increasing chain asymmetry and branching during reaction progression (Figure 4g). In contrast, both ratios remain nearly constant on Ir NP/WO_3, consistent with limited isomerization and predominant formation of linear alkane products.

Density functional theory (DFT) calculations were further performed to elucidate the distinct hydrogenolysis pathways of polyethylene fragments over Ir NP/WO_3 and Ir SA/WO_3. For Ir NP/WO_3 (Figure S32), C−C bond cleavage along the polymer backbone is associated with a relatively high kinetic barrier, whereas terminal C−C scission leading to methanation is thermodynamically favored. This energetic preference accounts for the low hydrogenolysis efficiency of polyolefins over Ir nanoparticles and the concomitant formation of substantial amounts of low-value methane. In contrast, over Ir SA/WO_3, the overall reaction landscape favors β-scission and skeletal isomerization pathways (Figure 4h). On the other hand, H_2 is dissociated at the isolated Ir site, followed by hydrogen spillover/transfer to adjacent oxygen sites, where the hydrogen species are stabilized. During C−C bond cleavage and skeletal rearrangement, these surface hydrogen species can be transferred back to hydrogen-deficient alkyl or carbocation-like intermediates, completing hydride reinsertion and final hydrogenation to saturated hydrocarbons. Along the β-scission route, β-hydride transfer is followed by sequential C−C bond cleavage, generating smaller alkyl fragments. The subsequent isomerization pathway proceeds with comparatively low energy barriers during alkyl migration, indicating that isolated Ir atoms effectively stabilize the β-hydride transfer transition state and facilitate hydride reinsertion, thereby providing sufficient geometric flexibility for carbon skeleton rearrangement and folding. This cooperative interaction couples C−C bond cleavage with chain rearrangement, thereby steering the reaction toward selective skeletal transformation rather than terminal cracking. Notably, due to the single-site adsorption configuration of polyethylene fragments on isolated Ir atoms, methanation of the terminal methyl group requires a substantially higher activation energy (1.61 eV), rendering it kinetically unfavorable relative to isomerization.

To further rationalize the pronounced difference in activation barriers, the projected density of states (PDOS) for the rate-determining steps in both methanation and isomerization pathways was analyzed. In the methanation pathway (Figure 4i), the occupied electronic states of the carbon backbone exhibit weak interaction with the vacant orbitals of the Ir center, necessitating traversal of a high-energy transition state. By contrast, in the isomerization pathway (Figure 4j), the occupied states of the carbon framework in the rate-determining configuration lie closer to the Fermi level and display stronger coupling with the vacant Ir orbitals. This enhanced electronic interaction lowers the activation barrier

Downloaded from pubs.acs.org/jacsat/article-pdf/148/31/33433/6330499/jacs.6c07894.pdf by INDIAN INST OF TECH DELHI- IIT user on 30 September 2026

https://doi.org/10.1021/jacs.6c07894
J. Am. Chem. Soc. 2026, 148, 33433−33442

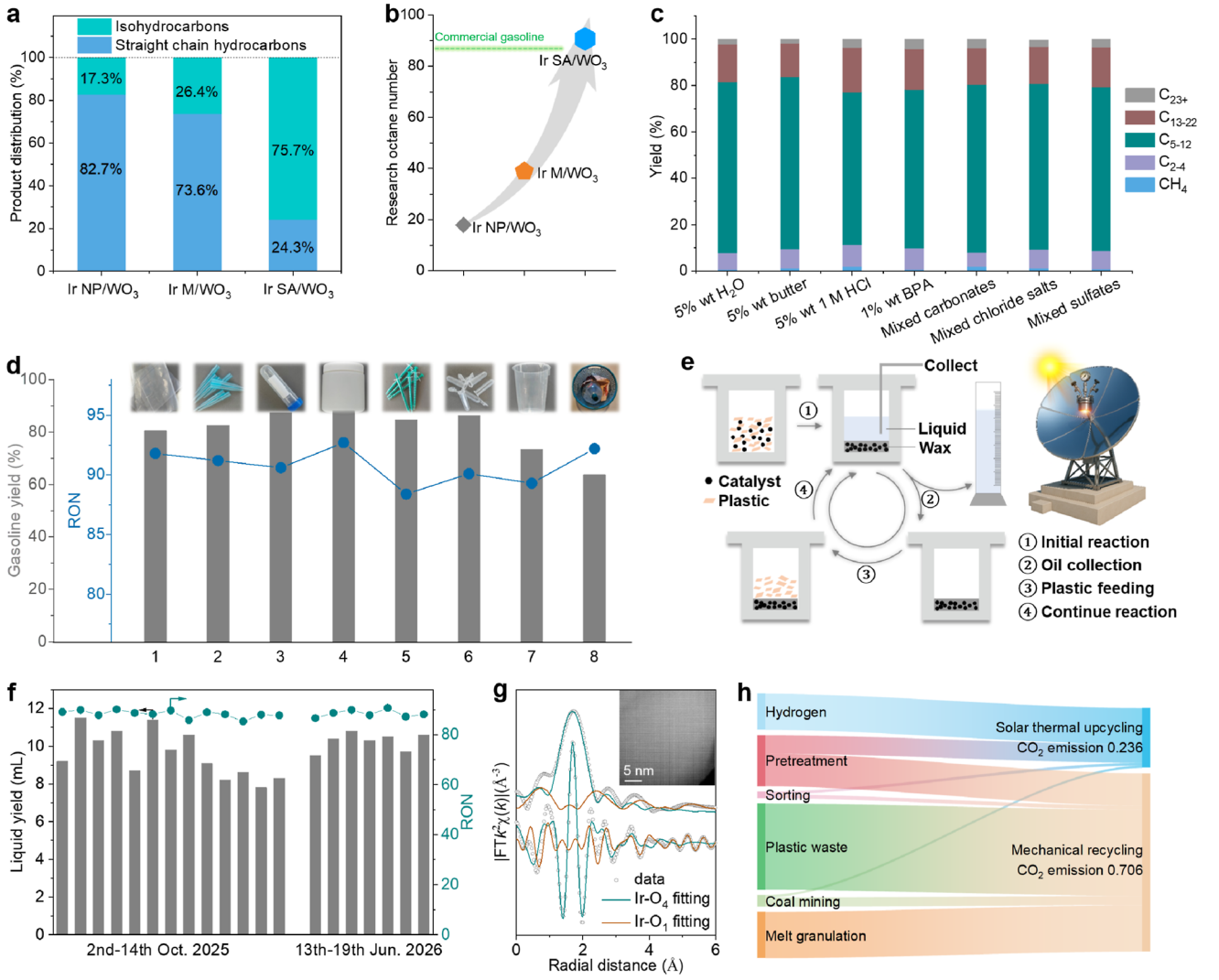


Figure 5. (a) The proportion of isomerized branched alkanes in liquid products. (b) Theoretical octane number of gasoline products obtained by different catalysts. (c) Catalytic performance of 1 g of LDPE (1 g of LDPE, 100 mg of catalyst, 20 bar H_2, 6 h outdoor reaction) with different additives (mixed carbonates including $CaCO_3$, Na_2CO_3, $MgCO_3$; mixed chloride salts including NaCl, KCl, $CaCl_2$; mixed sulfates including Na_2SO_4, $CaSO_4$, $MgSO_4$). (d) Upcycling performance of real plastic waste under outdoor solar thermal catalysis. (10 g of real plastic, 1 g of catalyst, 10 bar H_2, 6 h outdoor reaction). (e) Schematic diagram of long-term real-world plastic solar energy recycling. (f) Long-term continuous upcycling of real plastic waste. (g) Characterization of Ir SA/WO_3 catalysts after long-term use. (h) Sankey diagram of CO_2 emission analysis for upcycling and mechanical recycling processes.

significantly, thereby favoring skeletal isomerization over methane formation on isolated Ir sites.

As illustrated in Figure S33, these elementary steps collectively govern the transformation of long-chain alkanes (C_{12+}) into gasoline-range (C_{4-12}) branched isomers with high octane number and a minor fraction of diesel-range hydrocarbons. Notably, temperature-programmed reduction (H_2-TPR, Figure S34) reveals that Ir SA/WO_3 exhibits stronger hydrogen activation capacity compared to Ir NP/WO_3 and bare WO_3, as evidenced by the multiple hydrogen uptake events extending to higher temperatures. This enhanced dissociative H_2 activation is further confirmed by the H_2/D_2 exchange experiment (Figure S35), in which Ir SA/WO_3 displays a pronounced HD signal upon switching from H_2 to D_2, suggesting rapid hydrogen scrambling and spillover, underscores the pivotal role of atomically dispersed Ir sites in mediating H−H bond cleavage. This dual-pathway mechanism aligns with experimental observations from in situ DRIFTS and solid-state NMR, substantiating that atomically dispersed Ir centers uniquely balance hydrogenolysis and isomerization activities, in contrast to Ir nanoparticles that favor unselective linear chain cracking.

2.4. Practice of Solar Thermal Upcycling System

After elucidating the mechanism of polyolefin hydrogenolysis and isomerization over Ir SA/WO_3, we further quantified the proportion of isomerized alkanes in the gasoline-range products (Figures S36−S38). As illustrated in Figure 5a, the fraction of isomerized alkanes increased markedly from 17.3% to 75.7% as the Ir species evolved from nanoparticles to isolated single-atom sites, surpassing that of commercial gasoline (about 44.1%). Based on this composition, the calculated research octane number (RON) of the product meets the standard for premium commercial gasoline (Figure 5b).

We then evaluated the effects of moisture, grease contamination, acidic environments, common additives, and

https://doi.org/10.1021/jacs.6c07894
J. Am. Chem. Soc. 2026, 148, 33433−33442

Downloaded from pubs.acs.org/jacsat/article-pdf/148/31/33433/6330499/jacs.6c07894.pdf by INDIAN INST OF TECH DELHI- IIT user on 30 September 2026

various inorganic salts. As shown in Figure 5c, the addition of 5 wt % H_2O, butter, or 1 M HCl solution did not lead to any noticeable decrease in the hydrogenolysis activity of LDPE. Similarly, the introduction of antioxidants such as bisphenol A (BPA) exerted no significant influence on the catalytic performance. In addition, different inorganic salts acted merely as spectators in this system. These results indicate that typical impurities and contaminants present in household plastic wastes have negligible impact on the upcycling process.

After successfully verifying the catalytic applicability of Ir SA/WO_3 in various model systems, we further investigated the hydrogenolysis performance toward real polyolefin plastic wastes (Figures 5d and S39). Under standard outdoor solar-thermal conditions, a 6 h reaction enabled nearly complete conversion of various polyolefin plastics. In particular, plastic bags were fully transformed into liquid products accompanied by a small amount of gaseous products, while other high-rigidity polyolefin items retained a minor fraction of unconverted wax-like residues (Figure S40). Nevertheless, these waxy intermediates were not inert, they could be completely converted into liquid products upon a subsequent round of hydrogenolysis (Figure S41).

Based on these results, we designed a long-term continuous upcycling process for real plastic wastes (Figure 5e). In a typical experiment, 10 g of mixed plastic wastes and 1 g of catalyst were loaded into a high-pressure reactor. After each day's solar-thermal catalysis run, the liquid and gaseous products were collected and quantified, while the residual wax-catalyst mixture was left inside the reactor, supplemented with a fresh batch of plastic wastes for the next cycle. Through this operation, we conducted long-term plastic upgrading and recycling tests under representative outdoor conditions in both winter and summer (Figures 5f and S42). Despite variations in solar irradiation intensity and reactor temperature across different operation days, the system maintained stable daily conversion efficiency, reproducible product distributions, and high calculated RON values, demonstrating the durability of the Ir SA/WO_3 catalyst and the operational robustness of the solar-thermal upcycling process under realistic waste-to-fuel conditions (Figure 5g, Table S2, Figures S43 and S44).

More importantly, according to the life-cycle assessment (LCA) of carbon footprint (Figure S45 and Table S4), the proposed solar-thermal upcycling strategy for polyolefin plastics generates only 0.236 ton CO_2e per ton of feedstock, in which hydrogen production and supply account for the majority of the CO_2 emissions (Figure 5h). When green hydrogen is employed instead of conventional gray hydrogen, the greenhouse gas emissions can be further significantly reduced. In contrast, traditional mechanical recycling and incineration processes produce much higher carbon emissions, reaching 0.706 ton CO_2e per ton and 3.156 ton CO_2e per ton, respectively (Figure S46). Therefore, the proposed solar-thermal upcycling strategy can substantially reduce carbon emissions throughout the cradle-to-grave life cycle, contributing to the realization of a circular economy and the global carbon-neutrality goal.

3. CONCLUSIONS

In summary, this study demonstrates progress in the solar-thermal upcycling of polyolefin waste into renewable energy by reconstructing the reaction pathways of hydrogenolysis, enabling the direct production of high-octane gasoline from polyolefin waste. We illustrate that isolated Ir sites enforce a single-point adsorption configuration on β-scission intermediates along the polyolefin backbone, thereby providing sufficient geometric flexibility for carbon skeleton folding and rearrangement. This confinement significantly lowers the energy barrier for carbocation migration, leading to an unprecedented degree of skeletal isomerization and allowing the direct formation of highly branched gasoline-range hydrocarbons with research octane numbers exceeding 90. The integrated solar-thermal system enables continuous and sustainable transformation of real mixed plastic waste into high-quality gasoline, validating the practical feasibility of this approach. This work provides a viable pathway for transforming plastic waste into an "urban oilfield," while enhancing carbon efficiency and advancing the circular utilization of anthropogenic carbon resources.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/jacs.6c07894.

Experimental section and characterization details; XRD pattern; SEM images; TEM images, XPS spectra; NH_3-TPD; pyridine-adsorbed FTIR spectra; and activity comparison (PDF)

■ AUTHOR INFORMATION

Corresponding Author

Huabin Zhang − *Center for Renewable Energy and Storage Technologies (CREST), Physical Science and Engineering Division, King Abdullah University of Science and Technology, Thuwal 23955-6900, Kingdom of Saudi Arabia; KAUST Catalysis Center (KCC), Division of Physical Science and Engineering, King Abdullah University of Science and Technology (KAUST), Thuwal 23955-6900, Kingdom of Saudi Arabia;* orcid.org/0000-0003-1601-2471; Email: huabin.zhang@kaust.edu.sa

Authors

Chengyang Feng − *Center for Renewable Energy and Storage Technologies (CREST), Physical Science and Engineering Division, King Abdullah University of Science and Technology, Thuwal 23955-6900, Kingdom of Saudi Arabia; KAUST Catalysis Center (KCC), Division of Physical Science and Engineering, King Abdullah University of Science and Technology (KAUST), Thuwal 23955-6900, Kingdom of Saudi Arabia;* orcid.org/0000-0002-7437-1088

Miao Hu − *Center for Renewable Energy and Storage Technologies (CREST), Physical Science and Engineering Division, King Abdullah University of Science and Technology, Thuwal 23955-6900, Kingdom of Saudi Arabia; KAUST Catalysis Center (KCC), Division of Physical Science and Engineering, King Abdullah University of Science and Technology (KAUST), Thuwal 23955-6900, Kingdom of Saudi Arabia*

Shouwei Zuo − *Center for Renewable Energy and Storage Technologies (CREST), Physical Science and Engineering Division, King Abdullah University of Science and Technology, Thuwal 23955-6900, Kingdom of Saudi Arabia; KAUST Catalysis Center (KCC), Division of Physical Science and Engineering, King Abdullah University of Science and Technology (KAUST), Thuwal 23955-6900, Kingdom of Saudi Arabia*

Downloaded from pubs.acs.org/jacsat/article-pdf/148/31/33433/6633049/jacs.6c07894.pdf by INDIAN INST OF TECH DELHI- IIT user on 30 September 2026

Bo Li − *KAUST Catalysis Center (KCC), Division of Physical Science and Engineering, King Abdullah University of Science and Technology (KAUST), Thuwal 23955-6900, Kingdom of Saudi Arabia;* orcid.org/0000-0002-6179-4284

Yuanfu Ren − *Center for Renewable Energy and Storage Technologies (CREST), Physical Science and Engineering Division, King Abdullah University of Science and Technology, Thuwal 23955-6900, Kingdom of Saudi Arabia; KAUST Catalysis Center (KCC), Division of Physical Science and Engineering, King Abdullah University of Science and Technology (KAUST), Thuwal 23955-6900, Kingdom of Saudi Arabia*

Zhi-Peng Wu − *Center for Renewable Energy and Storage Technologies (CREST), Physical Science and Engineering Division, King Abdullah University of Science and Technology, Thuwal 23955-6900, Kingdom of Saudi Arabia; KAUST Catalysis Center (KCC), Division of Physical Science and Engineering, King Abdullah University of Science and Technology (KAUST), Thuwal 23955-6900, Kingdom of Saudi Arabia*

Jumanah Alharbi − *Center for Renewable Energy and Storage Technologies (CREST), Physical Science and Engineering Division, King Abdullah University of Science and Technology, Thuwal 23955-6900, Kingdom of Saudi Arabia; KAUST Catalysis Center (KCC), Division of Physical Science and Engineering, King Abdullah University of Science and Technology (KAUST), Thuwal 23955-6900, Kingdom of Saudi Arabia*

Jun Luo − *State Key Laboratory of Featured Metal Materials and Life-cycle Safety for Composite Structures, MOE Key Laboratory of New Processing Technology for Nonferrous Metals and Materials, School of Resources, Environment and Materials, Guangxi University, Nanning 530004, China;* orcid.org/0000-0003-2198-2421

Wan-Lu Li − *Aiiso Yufeng Li Family Department of Chemical and Nano Engineering, University of California, San Diego, La Jolla, California 92093-0448, United States;* orcid.org/0000-0003-0098-0670

Nikos Hadjichristidis − *KAUST Catalysis Center (KCC), Division of Physical Science and Engineering, King Abdullah University of Science and Technology (KAUST), Thuwal 23955-6900, Kingdom of Saudi Arabia;* orcid.org/0000-0003-1442-1714

Magnus Rueping − *KAUST Catalysis Center (KCC), Division of Physical Science and Engineering, King Abdullah University of Science and Technology (KAUST), Thuwal 23955-6900, Kingdom of Saudi Arabia;* orcid.org/0000-0003-4580-5227

Complete contact information is available at: https://pubs.acs.org/10.1021/jacs.6c07894

Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

This work was supported by the King Abdullah University of Science and Technology and the Center of Excellence for Renewable Energy and Storage Technologies under award number 5937, and the KAUST Supercomputing Laboratory under project K10175.

■ REFERENCES

(1) Borrelle, S. B.; Ringma, J.; Law, K. L.; Monnahan, C. C.; Lebreton, L.; McGivern, A.; Murphy, E.; Jambeck, J.; Leonard, G. H.; Hilleary, M. A.; et al. Predicted growth in plastic waste exceeds efforts to mitigate plastic pollution. *Science* **2020**, *369*, 1515−1518.

(2) Rochman, C. M.; Hoellein, T. The global odyssey of plastic pollution. *Science* **2020**, *368*, 1184−1185.

(3) MacLeod, M.; Arp, H. P. H.; Tekman, M. B.; Jahnke, A. The global threat from plastic pollution. *Science* **2021**, *373*, 61−65.

(4) Geyer, R.; Jambeck, J. R.; Law, K. L. Production, use, and fate of all plastics ever made. *Sci. Adv.* **2017**, *3*, No. e1700782.

(5) Ellis, L. D.; Rorrer, N. A.; Sullivan, K. P.; Otto, M.; McGeehan, J. E.; Román-Leshkov, Y.; Wierckx, N.; Beckham, G. T. Chemical and biological catalysis for plastics recycling and upcycling. *Nat. Catal.* **2021**, *4*, 539−556.

(6) Vora, N.; Christensen, P. R.; Demarteau, J.; Baral, N. R.; Keasling, J. D.; Helms, B. A.; Scown, C. D. Leveling the cost and carbon footprint of circular polymers that are chemically recycled to monomer. *Sci. Adv.* **2021**, *7*, No. eabf0187.

(7) Vogt, B. D.; Stokes, K. K.; Kumar, S. K. Why is recycling of postconsumer plastics so challenging? *ACS Appl. Polym. Mater.* **2021**, *3*, 4325−4346.

(8) Schyns, Z. O.; Shaver, M. P. Mechanical recycling of packaging plastics: a review. *Macromol. Rapid Commun.* **2021**, *42*, 2000415.

(9) Hu, M.; Feng, C.; Alharbi, J.; Zuo, S.; Ladelta, V.; Alshehry, G. S.; Ruwaili, K.; Al Qahtani, H. S.; Luan, D.; Li, W.-L.; et al. Isolated Ni atoms enable alkali-free photoreforming of waste polylactic acid plastic. *Nat. Commun.* **2026**, *17*, 347.

(10) Nicholson, S. R.; Rorrer, N. A.; Carpenter, A. C.; Beckham, G. T. Manufacturing energy and greenhouse gas emissions associated with plastics consumption. *Joule* **2021**, *5*, 673−686.

(11) Qiu, Z.; Lin, S.; Chen, Z.; Chen, A.; Zhou, Y.; Cao, X.; Wang, Y.; Lin, B.-L. A reusable, impurity-tolerant and noble metal−free catalyst for hydrocracking of waste polyolefins. *Sci. Adv.* **2023**, *9*, No. eadg5332.

(12) Kruger, D. D.; Cabrero-Antonino, M.; Osella, S.; Xu, F.; Yu, J.; Primo, A.; Garcia, H. Surface-State−Regulated Product Distribution in Photothermal CO_2 Hydrogenation Over MXene-Based S-Scheme Catalyst. *Angew. Chem., Int. Ed.* **2026**, *65*, No. e8425918.

(13) Zhang, F.; Zeng, M.; Yappert, R. D.; Sun, J.; Lee, Y.-H.; LaPointe, A. M.; Peters, B.; Abu-Omar, M. M.; Scott, S. L. Polyethylene upcycling to long-chain alkylaromatics by tandem hydrogenolysis/aromatization. *Science* **2020**, *370*, 437−441.

(14) Cen, Z.; Han, X.; Lin, L.; Yang, S.; Han, W.; Wen, W.; Yuan, W.; Dong, M.; Ma, Z.; Li, F.; et al. Upcycling of polyethylene to gasoline through a self-supplied hydrogen strategy in a layered self-pillared zeolite. *Nat. Chem.* **2024**, *16*, 871−880.

(15) Zhang, W.; Kim, S.; Wahl, L.; Khare, R.; Hale, L.; Hu, J.; Camaioni, D. M.; Gutiérrez, O. Y.; Liu, Y.; Lercher, J. A. Low-temperature upcycling of polyolefins into liquid alkanes via tandem cracking-alkylation. *Science* **2023**, *379*, 807−811.

(16) Xing, C.; Mao, C.; Wang, S.; Zhou, Y.; Wu, L.; Zhang, D.; Kang, D.; Yang, D.; Gong, W.; Wei, W.; et al. Ambient solar thermal catalysis for polyolefin upcycling using copper encapsulated in silicon nanosheets and chloroaluminate ionic liquid. *Nat. Catal.* **2025**, *8*, 556−568.

(17) Bi, T.; Chen, Y.; Lin, L.; Han, X.; Pan, Y.; Liu, C.; Cen, Z.; Luo, C.; Wen, W.; Zulfiqar, H.; et al. Closed-loop recycling of polyethylene to ethylene and propylene via a kinetic decoupling−recoupling strategy. *Nat. Chem. Eng.* **2025**, *2*, 650−661.

(18) Jia, X.; Qin, C.; Friedberger, T.; Guan, Z.; Huang, Z. Efficient and selective degradation of polyethylenes into liquid fuels and waxes under mild conditions. *Sci. Adv.* **2016**, *2*, No. e1501591.

(19) Wu, X.; Tennakoon, A.; Yappert, R.; Esveld, M.; Ferrandon, M. S.; Hackler, R. A.; LaPointe, A. M.; Heyden, A.; Delferro, M.; Peters, B.; et al. Size-controlled nanoparticles embedded in a mesoporous architecture leading to efficient and selective hydrogenolysis of polyolefins. *J. Am. Chem. Soc.* **2022**, *144*, 5323−5334.

Downloaded from pubs.acs.org/jacsat/article-pdf/148/31/33433/6330499/jacs.6c07894.pdf by INDIAN INST OF TECH DELHI- IIT user on 30 September 2026

(20) Zhao, J.; Hu, Q.; Wang, Y.; Che, Y.; Niu, Z. Tandem Catalysis to Mitigate Coke Formation in the Upcycling of Mixed Polyolefin Wastes. *J. Am. Chem. Soc.* **2025**, *147*, 30768−30776.
(21) Tian, S.; Bai, R.; Gao, Z.; Chen, Z.; Wang, M.; Tang, H.; Lin, S.; Xu, B.; Liu, X.; Yu, J.; et al. High-Efficiency Hydrocracking of Polyolefin Plastics by Controlling Intimacy between Pt Clusters and Zeolite Acid Sites. *J. Am. Chem. Soc.* **2025**, *147*, 30268−30276.
(22) Wang, M.; Gao, Y.; Yuan, S.; Deng, J.; Yang, J.; Yan, J.; Yu, S.; Xu, B.; Ma, D. Complete hydrogenolysis of mixed plastic wastes. *Nat. Chem. Eng.* **2024**, *1*, 376−384.
(23) Geng, Z.; Yu, Y.; Offen, A. J.; Liu, J. Achieving maximum overall light enhancement in plasmonic catalysis by combining thermal and non-thermal effects. *Nat. Catal.* **2023**, *6*, 1241−1247.
(24) Xu, F.; Zheng, L.; Zhang, J.; He, Y.; Cao, H.; Zheng, X.; García, H.; Yu, J. Co_3O_4 as full-solar-spectrum photocatalyst for selective methane conversion through reactive oxygen species control. *Nat. Catal.* **2026**, *9*, 73−86.
(25) Sharuddin, S. D. A.; Abnisa, F.; Daud, W. M. A. W.; Aroua, M. K. A review on pyrolysis of plastic wastes. *Energy Convers. Manage.* **2016**, *115*, 308−326.
(26) Kunwar, B.; Cheng, H.; Chandrashekaran, S. R.; Sharma, B. K. Plastics to fuel: a review. *Renew. Sust. Energy Rev.* **2016**, *54*, 421−428.
(27) Wang, C.; Xie, T.; Kots, P. A.; Vance, B. C.; Yu, K.; Kumar, P.; Fu, J.; Liu, S.; Tsilomelekis, G.; Stach, E. A.; et al. Polyethylene hydrogenolysis at mild conditions over ruthenium on tungstated zirconia. *JACS Au* **2021**, *1*, 1422−1434.
(28) Lee, W.-T.; van Muyden, A.; Bobbink, F. D.; Mensi, M. D.; Carullo, J. R.; Dyson, P. J. Mechanistic classification and benchmarking of polyolefin depolymerization over silica-alumina-based catalysts. *Nat. Commun.* **2022**, *13*, 4850.
(29) Lai, Q.; Zhang, X.; Jiang, S.; Krzyaniak, M. D.; Alayoglu, S.; Agarwal, A.; Liu, Y.; Edenfield, W. C.; Kobayashi, T.; Wang, Y.; et al. Stable single-site organonickel catalyst preferentially hydrogenolyses branched polyolefin C−C bonds. *Nat. Chem.* **2025**, *17*, 1488−1496.
(30) Zhou, Q.; Wang, D.; Wang, Q.; He, K.; Lim, K. H.; Yang, X.; Wang, W. J.; Li, B. G.; Liu, P. Mechanistic Understanding of Efficient Polyethylene Hydrocracking over Two-Dimensional Platinum-Anchored Tungsten Trioxide. *Angew. Chem., Int. Ed.* **2023**, *62*, No. e202305644.
(31) Zhang, Z.; Wang, J.; Ge, X.; Wang, S.; Li, A.; Li, R.; Shen, J.; Liang, X.; Gan, T.; Han, X.; et al. Mixed plastics wastes upcycling with high-stability single-atom Ru catalyst. *J. Am. Chem. Soc.* **2023**, *145*, 22836−22844.
(32) Chen, L.; Meyer, L. C.; Kovarik, L.; Meira, D.; Pereira-Hernandez, X. I.; Shi, H.; Khivantsev, K.; Gutierrez, O. Y.; Szanyi, J. Disordered, sub-nanometer Ru structures on CeO_2 are highly efficient and selective catalysts in polymer upcycling by hydrogenolysis. *ACS Catal.* **2022**, *12*, 4618−4627.
(33) Kots, P. A.; Xie, T.; Vance, B. C.; Quinn, C. M.; de Mello, M. D.; Boscoboinik, J. A.; Wang, C.; Kumar, P.; Stach, E. A.; Marinkovic, N. S.; et al. Electronic modulation of metal-support interactions improves polypropylene hydrogenolysis over ruthenium catalysts. *Nat. Commun.* **2022**, *13*, 5186.
(34) Chu, M.; Wang, X.; Wang, X.; Lou, X.; Zhang, C.; Cao, M.; Wang, L.; Li, Y.; Liu, S.; Sham, T.-K.; et al. Site-selective polyolefin hydrogenolysis on atomic Ru for methanation suppression and liquid fuel production. *Research* **2023**, *6*, 0032.
(35) Vance, B. C.; Najmi, S.; Kots, P. A.; Wang, C.; Jeon, S.; Stach, E. A.; Zakharov, D. N.; Marinkovic, N.; Ehrlich, S. N.; Ma, L.; et al. Structure−property relationships for nickel aluminate catalysts in polyethylene hydrogenolysis with low methane selectivity. *JACS Au* **2023**, *3*, 2156−2165.
(36) Dong, Q.; Lele, A. D.; Zhao, X.; Li, S.; Cheng, S.; Wang, Y.; Cui, M.; Guo, M.; Brozena, A. H.; Lin, Y.; et al. Depolymerization of plastics by means of electrified spatiotemporal heating. *Nature* **2023**, *616*, 488−494.

Downloaded from pubs.acs.org/jacsat/article-pdf/148/31/33433/6330499/jacs.6c07894.pdf by INDIAN INST OF TECH DELHI- IIT user on 30 September 2026

https://doi.org/10.1021/jacs.6c07894
J. Am. Chem. Soc. 2026, 148, 33433−33442